

Algorithm	Area	How it is used	Why it is used
Linear Regression	 House prices	Uses area, rooms, location, etc. to predict price	Output is a continuous number
Logistic Regression	 Loan approval	Uses income, age, credit history, etc. to predict Approve/Reject	Good for binary classification and gives probability
KNN	 Medical diagnosis	Compares a new patient's data with similar patients and uses their diagnosis	Useful when similar cases tend to have similar results
Decision Tree	 Banking	Makes decisions such as: income > X? → credit score > Y? → Approve/Reject	Easy to understand because it works like if-else rules
Random Forest	 Fraud detection	Many trees examine transaction features and vote Fraud/Normal	Multiple trees usually give more robust predictions than one tree
SVM	 Spam detection	Converts email features into data points and finds a boundary separating spam/non-spam	Effective when classes can be separated by a clear boundary
Naive Bayes	 Text classification	Checks words/features in a text and calculates probability of each category	Fast and effective for text data

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Gradient Boosting	 Customer churn	Predicts whether a customer will leave; each new tree focuses on previous mistakes	Good for learning complex relationships in tabular data
Neural Network	 Image recognition	Learns patterns such as edges → shapes → objects from images	Can learn very complex patterns automatically